

Data and Network Security

(Master Degree in Computer Science and Cybersecurity)

Lecture 13
Spring 2026



Outline for today

- **Social Bots**
- **Detection of Spambot Groups**

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Social Bots - Spam bots

(Semi-)automated accounts with usually harmful intention designed to mimic human behavior on social media platforms.

Usual intentions:

- Misinformation spreading
- Stealing personal information/data
- Stock market manipulation
- Political influence

Social Bots - some history

- Started with fake followers

	1000	2500	5000	10 000	25 000	50 000
Buy Twitter Followers	\$20	\$40	\$70	\$160	\$300	\$500
Twitter Followers	1000	2500	5000	10 000	25 000	50 000
Delivery Time (days)	2-3	2-3	2-3	2-3	2-3	3-5
Money-Back Guarantee	✓	✓	✓	✓	✓	✓
	Buy Now	Buy Now	Buy Now	Buy Now	Buy Now	Buy Now

Social Bots - current spambots

- Nowadays we have social spambots
 - Aim: Indistinguishable from genuine accounts

They can be:

- follower bots,
- like bots,
- comment bots,
- content-sharing bots
- ...

Social Bots - what they usually do/affect?

- Distortion of online discussion
- Manipulation of overall public opinion
- Lowering trust levels
- Misinformation amplification
- Possible economic implications
- Psychological implications

Social Bots - what they usually do/affect?

- Distortion of online discussion

By typically flooding platforms with repetitive or misleading content, spambots can push over the audiences certain viewpoints or narratives, artificially inflating their popularity or significance.

This distortion can make it difficult for users to discern genuine opinions and information from manipulated or fabricated content.

Social Bots - what they usually do/affect?

- Manipulation of overall public opinion

Engaging in coordinated campaigns to promote or discredit certain viewpoints, products, or political candidates this artificially generated content can alter public perceptions and attitudes, potentially impacting real-world outcomes such as elections or consumer behavior.

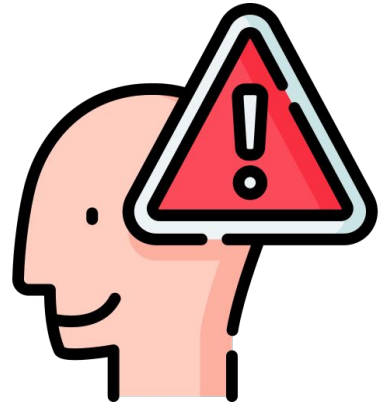


Social Bots - what they usually do/affect?

- Lowering trust levels

Aware users may become skeptical of the authenticity of social media engagement, leading to decreased trust in platforms and their content.

Fake accounts and automated interactions can tarnish the reputation of legitimate users and organizations, further eroding trust within online communities.



Social Bots - what they usually do/affect?

- Economic implications

Influencing consumer behavior and online market dynamics.

Spambots can inflate metrics such as:

- likes, shares, and followers,

This can mislead advertisers/investors about the true reach and engagement of content of a “celebrity” which down the line with result in financial loss.

Social Bots - what they usually do/affect?

- Psychological implications

Exposure to manipulated or deceptive content may disrupt individuals' confidence/judgement in their ability to discern truth from false.



Outline for today

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- Detection of Spambot Groups

Social Bots - How to detect them?



Social Bots - How to detect them?

Nowadays spambots have become indistinguishable from genuine accounts if analyzed one-by-one so:

Social Bots - How to detect them?

Nowadays spambots have become indistinguishable from genuine accounts if analyzed one-by-one so:

How about analyzing the online behavior of large groups of users, with the goal of detecting possible spambots among them?

Behavioral analysis

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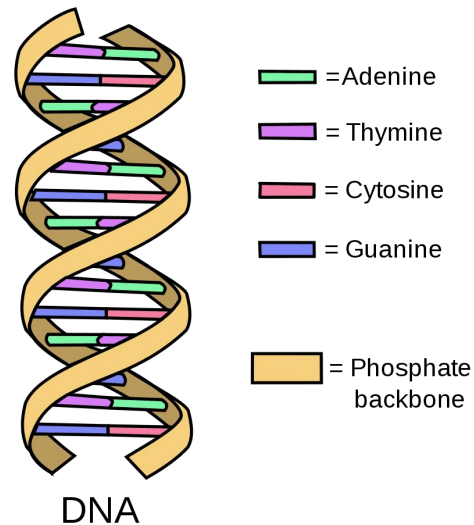
Behaviour – sequence of actions performed by an account

Behavioral analysis

Behaviour – sequence of actions performed by an account.

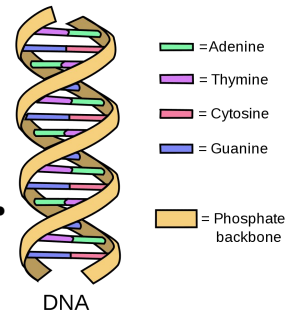
Inspiration from DNA:

- Each type of action is associated to a character (e.g., A, B, C)
- The online behaviour of an account is modeled as a sequence of characters according to the sequence of actions performed by that account



Behavioral analysis

Behaviour – sequence of actions performed by an account.



By drawing a parallel with biological DNA, we will see how to model users behaviors and interactions by means of strings of characters, representing the sequence of their actions.

Online actions—such as posting new content, replying to another user, following an account can be encoded with different characters, similarly to DNA sequences.

Digital DNA

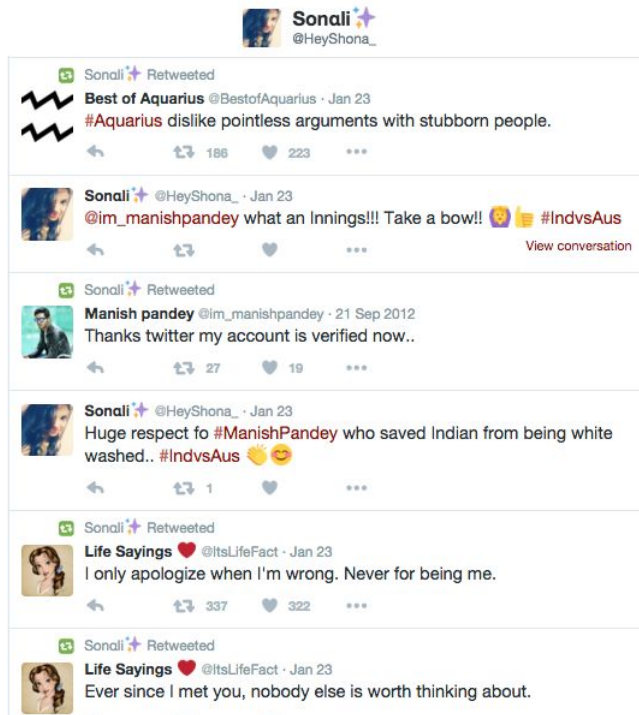


Encoding

T – tweet

R – retweet

P – reply



R

P

R

T

R

R

RPRTRR

Digital DNA vs biological DNA



T - tweet
R - retweet
P - reply

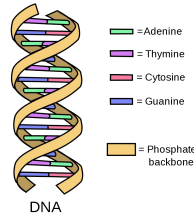
...RRTRPRTPRRPRTPRPTPRRTRPR
...RPRTPTTRPTRPTPRRRRTPPRPP
...TTTRRRPPTPRPTPRTRPTRRRTP
...PRTRPRTPPPPRTPRRPRTPPRRT
...TRTRPRTPRRPRTPRPTPTPPRTT
...TRPPRTPPTRPPTPRRTTTPPRPR

A - adenine
G - guanine
T - thymine
C - cytosine

...AGTCTCCATTTTCAGGTCGTA
...GTTTAAGATCGCCTCATCACC
...AGGCAATTCGCCTGAACTGG
...AGTCTCGATCCTTTCCTCGTT
...AAAATCGAACGCCTTGTCGG
...ATTCTCCATCGCCTAAACAAC

Spambot characterization

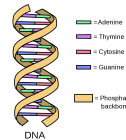
Automated accounts have similar DNA sequences.



The most well-known and widely adopted analysis techniques are **sequence alignment** and **repetition/pattern elicitation**. One of the main goals of these techniques is to find commonalities and repetitions among DNA sequences.

Via an analysis of common sub-sequences and substrings it is possible to predict specific characteristics of the individual and to uncover relationships between different individuals.

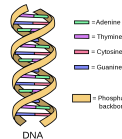
Spambot characterization



To create a digital DNA sequence all the possible actions are represented as a finite set of unique characters:

$$\mathbb{B} = \{B_1, B_2, \dots, B_N\} \quad B_i \neq B_j \quad \forall \quad i, j = 1, \dots, N \quad \wedge \quad i \neq j.$$

Spambot characterization



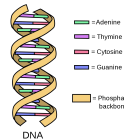
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A digital DNA sequence is an ordered tuple, or vector, of characters whose possible values are defined by the bases of its alphabet. A sequence s is defined as:

$$s = (b_1, b_2, \dots, b_n) \quad b_i \in \mathbb{B} \quad \forall \quad i = 1, \dots, n.$$

Spambot characterization

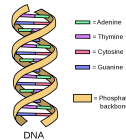


Different alphabets can model different granularities

$$\begin{aligned}\mathbb{B}_{content}^3 &= \left\{ \begin{array}{l} \text{N} \longleftarrow \text{tweet contains no entities (plain text),} \\ \text{E} \longleftarrow \text{tweet contains entities of one type,} \\ \text{X} \longleftarrow \text{tweet contains entities of mixed types} \end{array} \right\} \\ &= \{\text{N}, \text{E}, \text{X}\}\end{aligned}$$

$$\begin{aligned}\mathbb{B}_{content}^6 &= \left\{ \begin{array}{l} \text{N} \longleftarrow \text{tweet contains no entities (plain text),} \\ \text{U} \longleftarrow \text{tweet contains one or more URLs,} \\ \text{H} \longleftarrow \text{tweet contains one or more hashtags,} \\ \text{M} \longleftarrow \text{tweet contains one or more mentions,} \\ \text{D} \longleftarrow \text{tweet contains one or more medias,} \\ \text{X} \longleftarrow \text{tweet contains entities of mixed types} \end{array} \right\} \\ &= \{\text{N}, \text{U}, \text{H}, \text{M}, \text{D}, \text{X}\}.\end{aligned}$$

Longest common substring

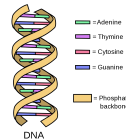


Observe the Longest substring between N sequences of digital DNA

```
...TRRRPRRTRRPRTPRPTPRRTRPR  
...RPRTPTTRRRPRRTPRRRRTPPRP  
...TTTRRRPRRRPRRTRTRPTRRRTP  
...PRTRPRTPPPPRTPRRRRRPRRTR
```

Intuitively, users that share long behavioral patterns are much more likely to be similar than those that share little to no behavioral patterns.

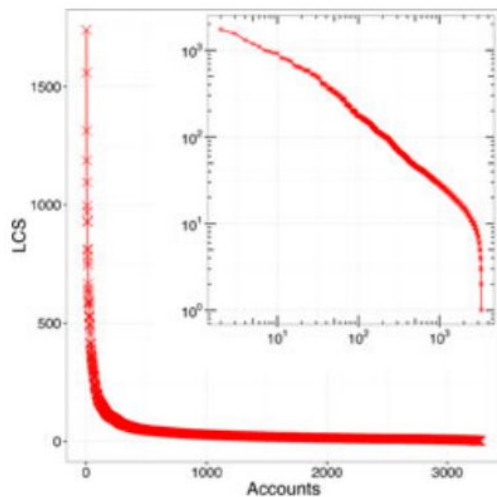
Longest common substring



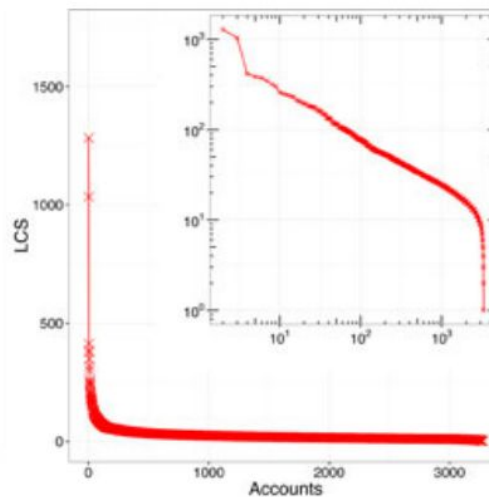
Goal: find the LCS that is common to at least k of these strings:

$$A = \begin{pmatrix} s_1 \\ s_2 \\ \vdots \\ s_M \end{pmatrix} = \begin{pmatrix} (b_{1,1}, b_{1,2}, \dots, b_{1,n}) \\ (b_{2,1}, b_{2,2}, \dots, b_{2,m}) \\ \vdots \\ (b_{M,1}, b_{M,2}, \dots, b_{M,p}) \end{pmatrix}$$

Longest common substring



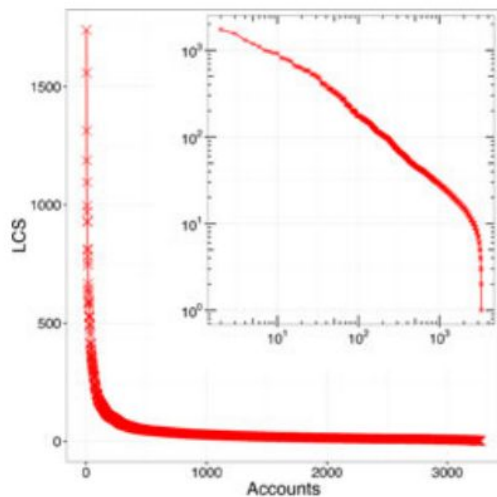
(a) $\mathbb{B}_{type}^3$ alphabet.



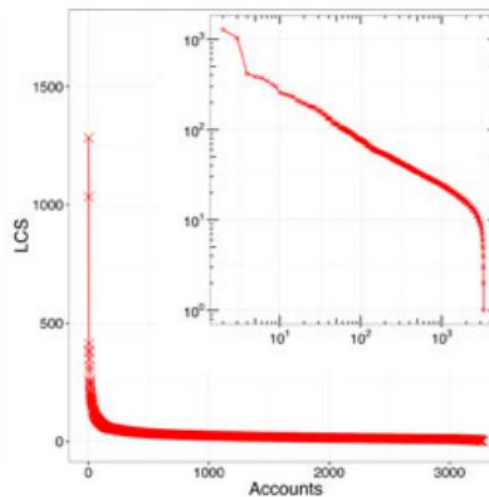
(b) $\mathbb{B}_{content}^3$ alphabet.

- x axis - the number of k accounts (corresponding to the k strings, used to compute LCS values)
- y axis - the length of the LCS common to at least k accounts.

Longest common substring



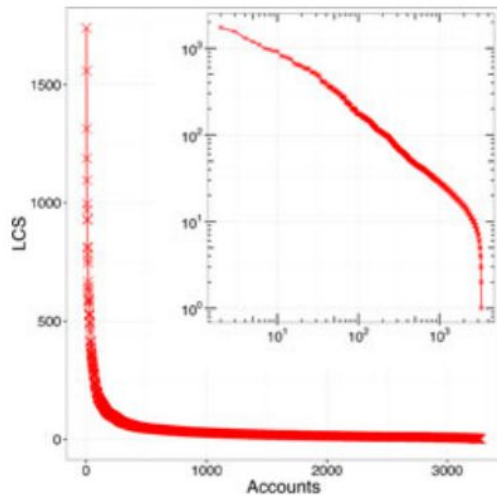
(a) $\mathbb{B}_{type}^3$ alphabet.



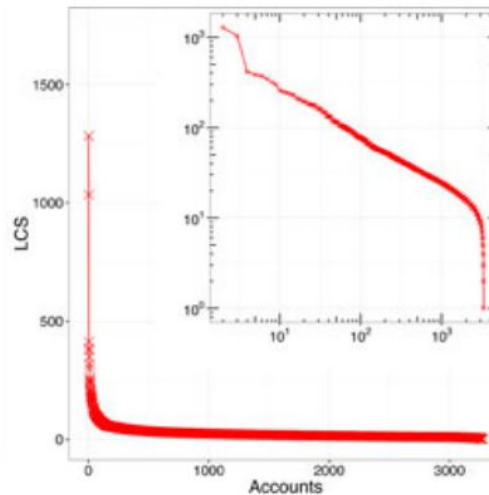
(b) $\mathbb{B}_{content}^3$ alphabet.

Each point in an LCS curve corresponds to a subset of k accounts that share the longest substring (of length y) among all those shared between all the other possible subsets of k accounts.

Longest common substring



(a) $\mathbb{B}_{type}^3$ alphabet.

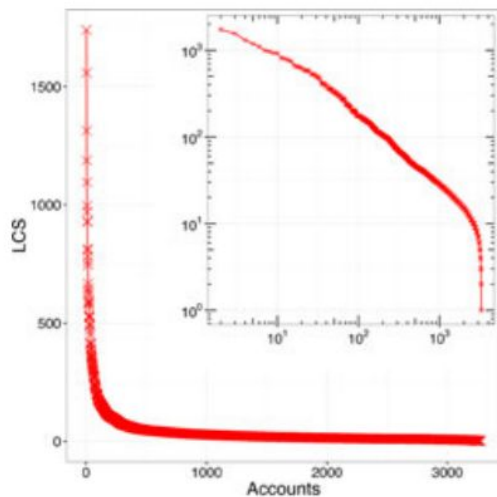


(b) $\mathbb{B}_{content}^3$ alphabet.

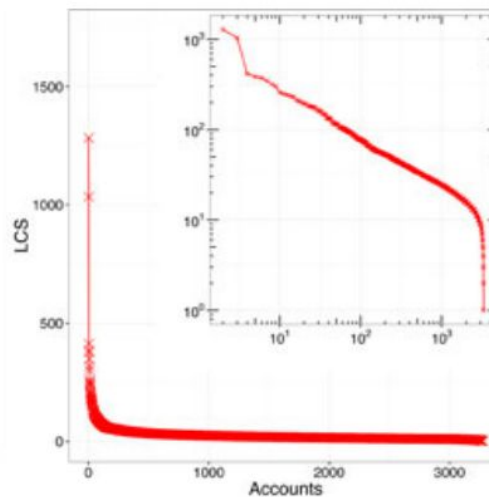
As the number k of accounts grows,
the length of the LCS common to all of them shortens.

LCS curves are monotonic non-increasing functions.

Longest common substring



(a) $\mathbb{B}_{type}^3$ alphabet.

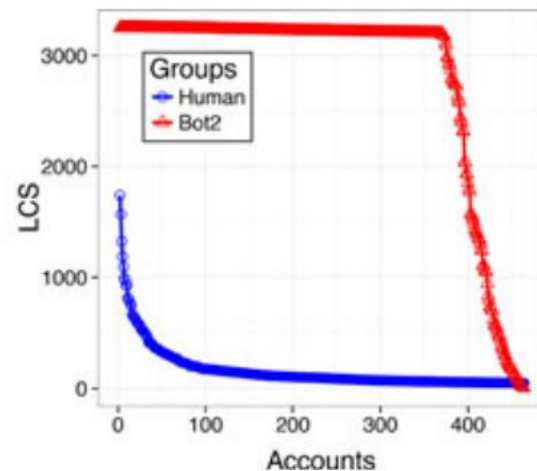
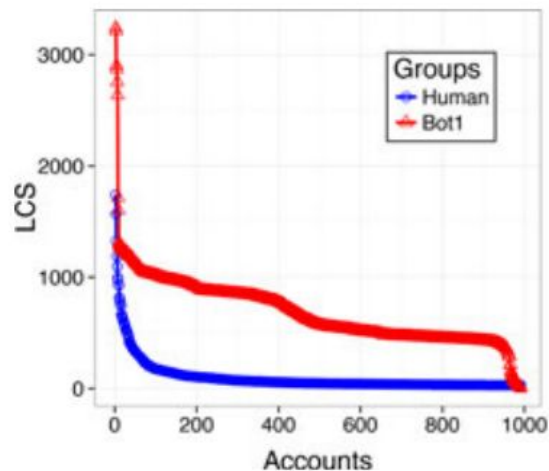


(b) $\mathbb{B}_{content}^3$ alphabet.

Thus, it is more likely to find a long LCS among a few accounts rather than among large groups.

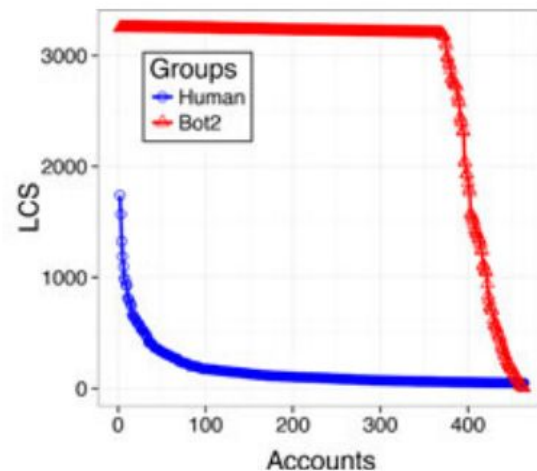
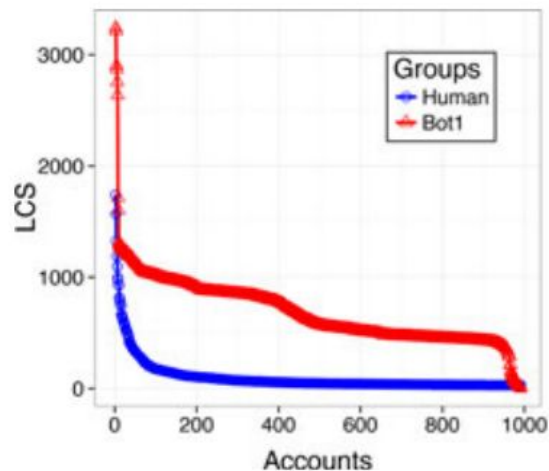
LCS curves are monotonic non-increasing functions.

Bots versus Humans



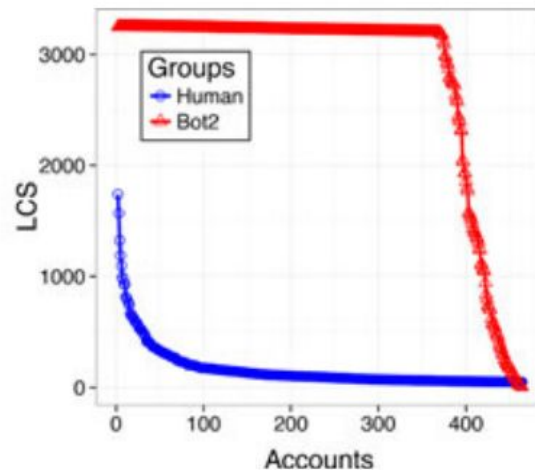
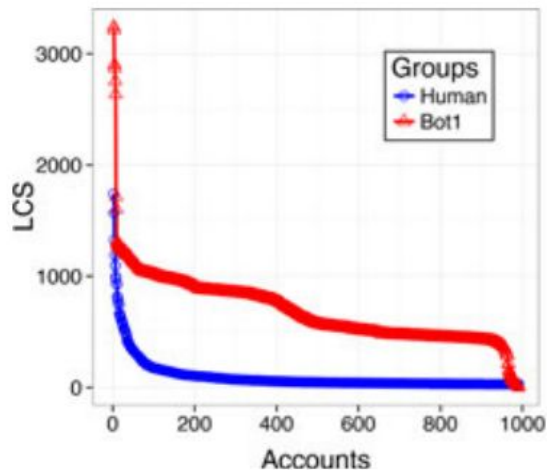
Observation 1: LCS of both groups of spambots are rather long even when the number of accounts grows.

Bots versus Humans



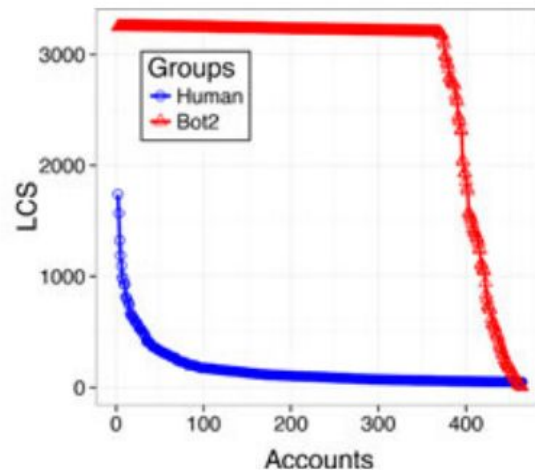
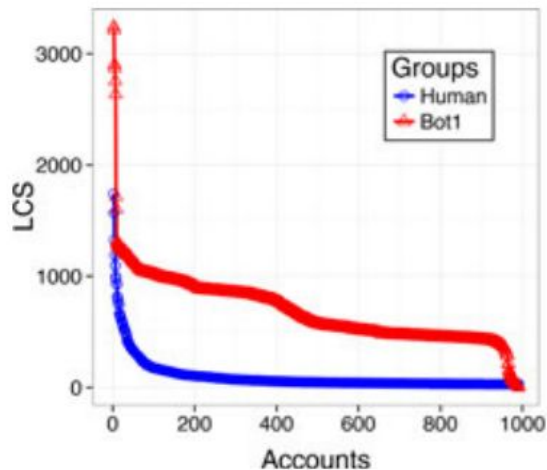
Observation 2: Sudden drop of the LCS length when the number of accounts gets close to the group size for spambots

Bots versus Humans



Observation 3: Genuine accounts show little to no similarity-as represented by LCS curves that exponentially decay, rapidly reaching the smallest values of LCS length.

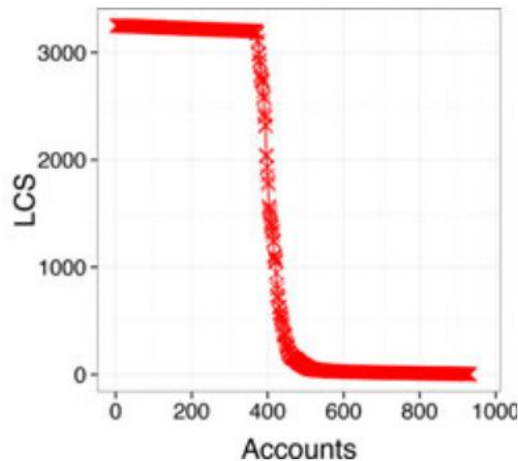
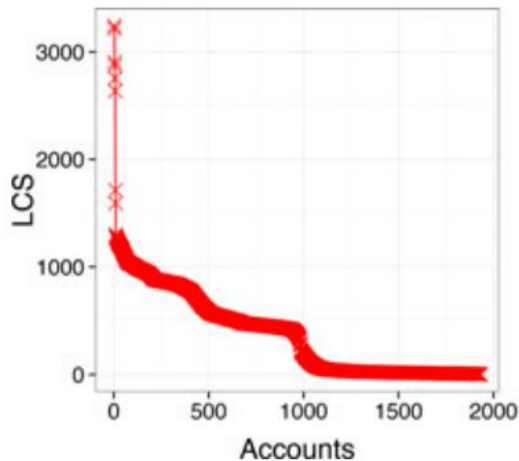
Bots versus Humans - Automation in distinguishing?



Consider high behavioral similarity as a proxy for automation and, thus, a high level of similarity among a large group of accounts might serve as an indicator for anomalous behaviors.

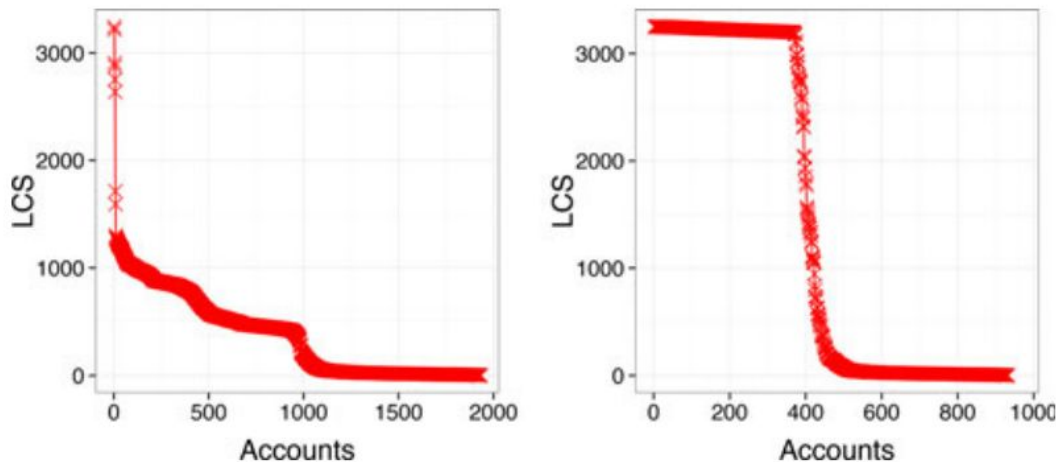
Heterogeneous Users - LCS

Mix various bot families and same quantity of normal users data.



Heterogeneous Users - LCS

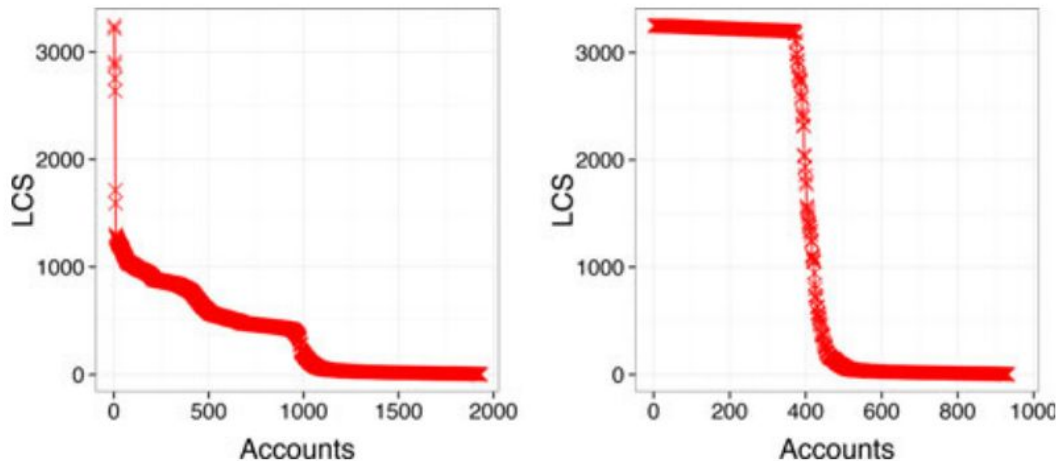
Mix various bot families and same quantity of normal users data.



LCS curves in both plots asymptotically reach their minimum value as the number of accounts grows.

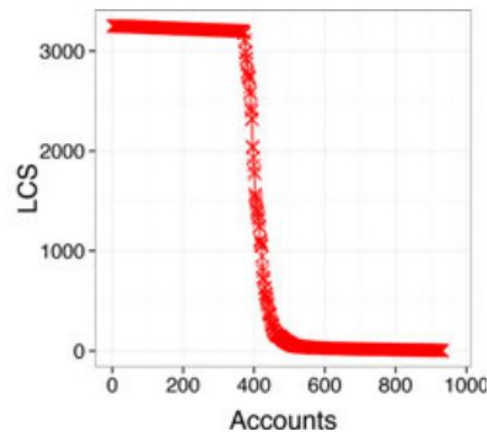
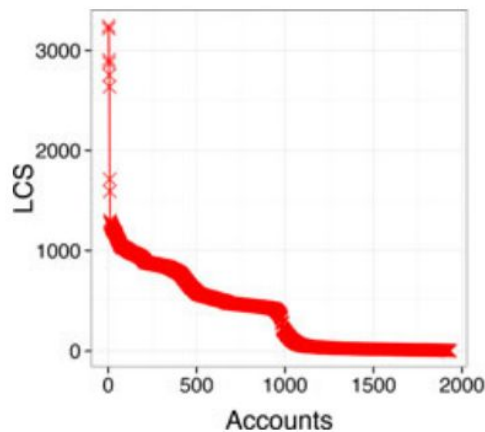
Heterogeneous Users - LCS

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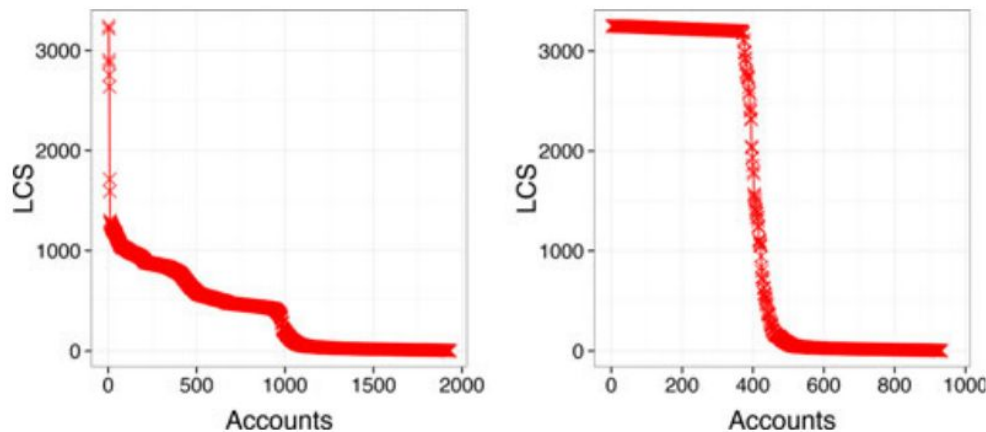
We have a different behaviour compared to the individual groups

Heterogeneous Users - LCS



Observation: A trend seems to be dominant only until reaching a certain threshold. Then, a steep fall occurs and another possibly different-trend kicks in.

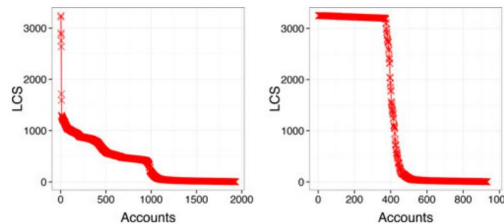
Heterogeneous Users - LCS



Observation: A trend seems to be dominant only until reaching a certain threshold. Then, a steep fall occurs and another possibly different-trend kicks in.

Observation: These portions of the LCS curves separated by the steep drops resemble LCS curves of the single groups of similar users

Heterogeneous Users - LCS



The steep drops of LCS curves separate areas where the length of the LCS remains practically unchanged, even for significantly different numbers of considered accounts.

These plateaux in LCS curves are strictly related to homogeneous groups of highly similar accounts. (multiple plateaux - multiple sub groups existing).

The steeper the drop in a LCS curve, the more different are the two subgroups of accounts split by that drop.

Slow and gradual decreases in LCS curves represent areas of uncertainty, where it might be difficult to make strong hypotheses about the characteristics of the underlying accounts.

Time for detection

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Supervised approach - case 1

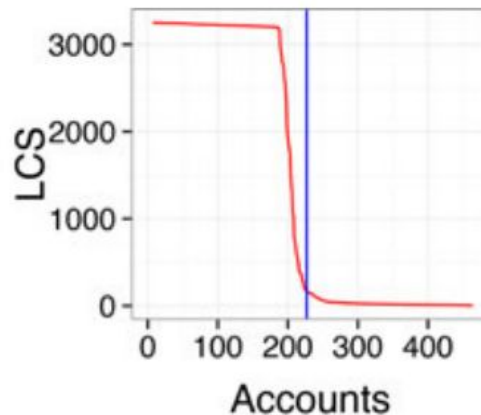
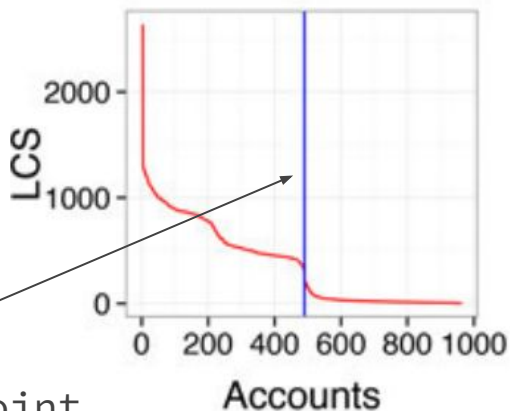


A good division of the original set of users into several subgroups is one where all the users belonging to a given class are assigned to the same subgroup.

LCS curve of a heterogeneous group of users can be used as a splitting point to obtain two subgroups of more homogeneous users.

Supervised approach - case 1 - evaluation

Using a labeled dataset check all possible splitting point in the LCS curve of the training-set users and find the one that yields the best possible subgroup division. (Every point generates a different “classifier”)

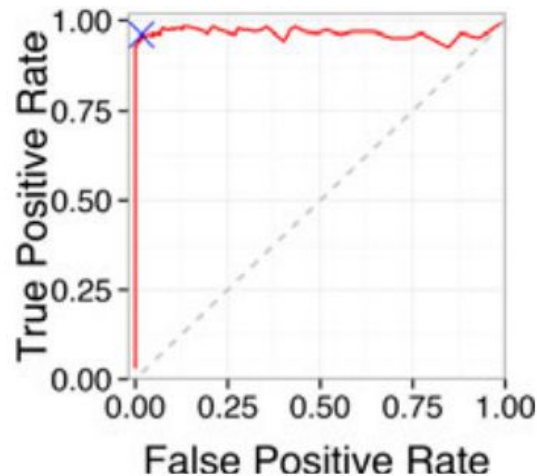
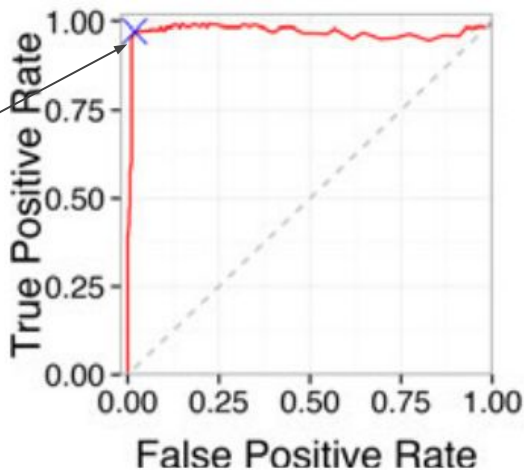


Best splitting point
Left - spambots
Right - genuine users

Supervised approach - case 1 - evaluation

Supervised - Cannot guarantee that the learned LCS value would still be effective when applied on a test-set different from the one used to derive the LCS value.

Best model



Unsupervised approach

Exploit the discrete derivative of a LCS curve to recognize the points corresponding to the steep drops.

Unsupervised approach

Exploit the discrete derivative of a LCS curve to recognize the points corresponding to the steep drops.

The steep drops of LCS curves appear as sharp peaks in the derivative plot

- represent suitable splitting points to isolate different subgroups among the whole set of users.

Unsupervised approach

Exploit the discrete derivative of a LCS curve to recognize the points corresponding to the steep drops.

The steep drops of LCS curves appear as sharp peaks in the derivative plot

- represent suitable splitting points to isolate different subgroups among the whole set of users.

Rank the suitable points according to their corresponding derivative value

- how steep is the corresponding drop

Unsupervised approach

Exploit the discrete derivative of a LCS curve to recognize the points corresponding to the steep drops.

The steep drops of LCS curves appear as sharp peaks in the derivative plot

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Rank the suitable points according to their corresponding derivative value

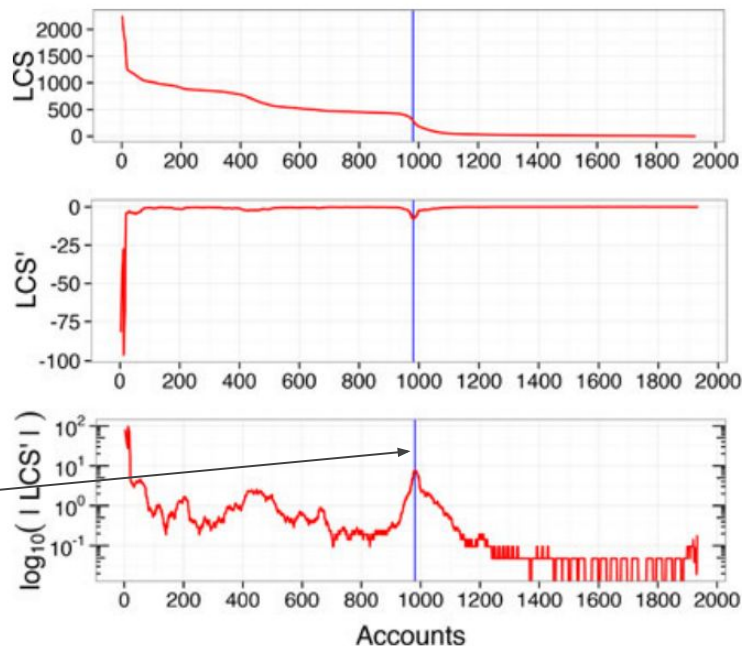
- how steep is the corresponding drop

Repeatedly divide the whole set of users based on the ranked points, leading to a dendrogram structure.

- Useful when the LCS curve exhibits multiple plateaux and steep drops, (find best possible clusters)

Unsupervised approach

Exploit the discrete derivative of a LCS curve to recognize the points corresponding to the steep drops.



Most pronounced
peak

Reading Material

1. Social Spambot detection: [Link-1](#), [Link-2](#), [Link-3](#)